

TECHNICAL SPECIFICATION DOCUMENT

MediBook — Patient Portal & Clinic Management System

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This document describes the proposed technical architecture, integration approach, security design, and infrastructure requirements for the MediBook platform. It is intended for review by the Client's IT team and is subject to revision following confirmation of open items noted in the BRD.

1. System Architecture Overview

MediBook is designed as a three-tier web application with a separately deployed native mobile client. The architecture is cloud-hosted and stateless at the application layer to support horizontal scalability.

Layer	Technology	Notes
Frontend (web)	React.js (TypeScript)	Responsive; tested on Chrome, Firefox, Safari (latest 2 versions)
Backend API	Node.js + Express.js	RESTful API; JWT-based authentication and authorisation
Database	PostgreSQL 15	Primary relational store; structured patient and clinical data
Mobile app	React Native	Single codebase targeting Android and iOS; native build per platform
File storage	AWS S3 or equivalent	Prescriptions, lab reports, profile photos; versioned, encrypted
Session / cache	Redis	Session management, appointment slot cache, search index
Cloud hosting	TBD — Client to confirm	AWS, Azure, or GCP; India region mandatory for patient data residency

Note: Cloud provider choice requires Client confirmation (see BRD OQ-06). Data residency in India is non-negotiable under applicable Indian healthcare data laws.

2. Integration Specifications

2.1 Medi-Plus v3.2 (Legacy EMR)

Read-only integration to retrieve historical patient records for display in the doctor dashboard.

Item	Detail
Integration type	REST API preferred; database-level read (JDBC/ODBC) as fallback if API unavailable
Data consumed	Patient demographics, past visit summaries, historical prescription records
Access method	PENDING — API documentation not yet received from Client IT team
Authentication	API key or service account credentials to be provided by Client
Fetch strategy	On-demand fetch when doctor opens a patient record (real-time, not batch)
Fallback behaviour	If API unavailable, display an alert banner; serve MediBook-native records only

2.2 Tally ERP (Billing System)

Bidirectional sync of invoice and payment status between MediBook and the Client's Tally instance.

Item	Detail
Integration type	Tally Prime REST API if available; XML file-based export/import as fallback
MediBook → Tally	Push new invoice record on visit completion
Tally → MediBook	Sync payment status (paid / partially paid / outstanding)
Sync frequency	Invoice push: near real-time. Payment status: polled every 5 minutes.
Confirmation	PENDING — Client IT to advise on available Tally connector version and setup

2.3 Payment Gateway — Razorpay

Client confirmed Razorpay verbally during discovery call (15 Dec 2025). Written confirmation and credentials are outstanding per BRD OQ-01.

Item	Detail
Supported methods	UPI, Netbanking, Debit/Credit cards, Wallets (standard Razorpay methods)
Integration type	Razorpay server-side Node.js SDK + client-side Checkout widget
Webhook	Payment status webhook from Razorpay to MediBook backend (HMAC verified)
Refunds	Triggered from admin panel; processed via Razorpay Refund API
Credentials needed	Razorpay API Key ID and Key Secret — to be provided by Client

2.4 WhatsApp Business API

Required for automated appointment reminders (BRD requirements NOT-001 and NOT-002). Requires an approved WhatsApp Business Account with Meta.

Item	Detail
Provider	Meta Cloud API preferred; BSP alternatives: Twilio, Gupshup, or Kaleyra
Message templates	Pre-approved transactional templates required before sending (Meta approval process ~2–5 days)
Status	PENDING — Client IT to confirm if account exists or fresh provisioning needed (BRD OQ-04)
Fallback	SMS-only notifications if WhatsApp API not available at Phase 3 delivery

3. Security and Compliance

Requirement	Implementation approach
Encryption in transit	TLS 1.3 for all API and web traffic; HTTPS enforced with HSTS headers
Encryption at rest	AES-256 for database storage volumes and S3 file buckets (SSE-S3 or SSE-KMS)
Authentication	OTP-based login for patients; username + TOTP-MFA for all clinic staff
Authorisation	RBAC with five roles: Patient, Doctor, Receptionist, Admin, Super-Admin
Audit logging	All PHI access events logged: user ID, timestamp, resource accessed, action taken
Data residency	All data hosted in India-region cloud nodes; zero cross-border data transfer
OWASP coverage	OWASP Top 10 addressed in code review checklist; quarterly pen testing recommended
Compliance framework	HIPAA-aligned practices applied; applicable Indian law is DPDP Act 2023 and ABDM guidelines

Note: The BRD references 'HIPAA-aligned' practices. MediBook operates under Indian jurisdiction. The precise applicable framework is the Digital Personal Data Protection (DPDP) Act, 2023 and ABDM health data guidelines. Client's legal team should confirm exact compliance obligations before Phase 1 architecture sign-off.

4. Performance Requirements

Specific SLAs are under discussion with the Client (BRD OQ-06). The following are the Vendor's proposed baseline targets, subject to written confirmation.

Metric	Proposed target	Basis
Doctor dashboard page load	< 3 seconds at P95	Per Client email request (Jan 12)
API response (booking, Rx)	< 1 second at P95	Excluding third-party latency (EMR, payment)
Monthly uptime	99.5%	Excludes pre-notified maintenance windows
Peak concurrent users	~25 staff + up to 200 patient sessions	Based on discovery call data (15 Dec)

Metric	Proposed target	Basis
File upload	Up to 10 MB per lab report or PDF	
Load testing	Conducted in Phase 4 before go-live	Results may trigger architecture changes

Note: Performance targets are not contractually binding until confirmed in writing. Load testing (Phase 4) will determine whether targets are met and may require infrastructure scaling adjustments.

5. Data Migration

Existing patient records from Medi-Plus v3.2 will be migrated to MediBook as a one-time operation during the Phase 4 go-live window.

Item	Status / detail
Record count	Approximately 40,000 patient records (per Client, discovery call 15 Dec 2025)
Source format	Medi-Plus CSV export — sample file expected from Client IT by 13 Jan 2026
Target schema	MediBook patient and visit tables (schema to be shared in Phase 1 deliverable)
Migration method	ETL script developed by Stackular; dry-run on staging env before production cutover
Validation	Client to review a 10% stratified sample of migrated records for accuracy sign-off
Access post-migration	Historical records available read-only via the doctor dashboard
Rollback plan	Medi-Plus remains active and writable until post-migration validation is complete

Note: Data volume and format are provisional pending receipt of a sample export file from Client IT. Significant schema deviations from the expected Medi-Plus CSV structure may impact Phase 3 timeline.